

ON SOLID ANGLES AND SIMPLEX SHAPE MEASURES

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ABSTRACT. The minimum solid angle of a simplex is shown to be a shape measure in any dimension. While this has been known in dimensions two and three, the higher-dimensional generalization is new. We prove inequalities between the minimum solid angle of a non-degenerate simplex and its fullness. Up to dimension-dependent constants, the fullness is bounded from above by the minimum solid angle and from below by the square of the minimum solid angle. The exponents are optimal for three and higher dimensions. This work therefore generalizes the minimum angle condition. We also review several simplex shape measures and their relationships.

1. INTRODUCTION

How to measure the quality of triangulations has been studied extensively throughout the mathematical and quantitative sciences. Intuitively, some triangles and tetrahedra are more *regular* or *well-shaped* than others. *Shape measures* formalize and quantify the quality of such shapes. Examples of such shape measures include the *aspect ratio* and the *fullness* of a simplex. Beyond triangles and tetrahedra, shape quality indicators evaluate simplices in higher dimensions. In this note, we show that the *minimum solid angle* is a simplicial shape measure, polynomially comparable to the aspect ratio and the fullness. While this has been known for triangles and tetrahedra, the generalization to simplices of arbitrary dimension seems to have been absent from the literature up to now.

Notions of shape measures appear throughout the mathematical literature. Shape measures associate a real number to a simplex, continuous in the coordinates of the vertices and invariant under reflection, rotation, translation, and positive scaling. They are also independent of the order of the vertices. For example, an n -dimensional simplex S with diameter δ_S , volume V_S , and inradius r_S has the *aspect ratio* $\alpha_S := r_S/\delta_S$ and the *fullness* $\mu_S := V_S/\delta_S^n$. Clearly, the aspect ratio and the fullness are invariant under orthogonal transformations, translations, and positive rescaling of simplices. If the simplex shows any type of degeneracy, then the shape measures deteriorate, but if the simplex has a regular shape, then the shape measures remain at values close to the optimum.

In two dimensions, the minimum angle $\gamma_{\min}$ of a triangle S is a shape measure. It is easy to verify that this shape measure is equivalent to the fullness, in the sense that

$$\frac{1}{4}\gamma_{\min} \leq \mu_S \leq \frac{1}{2}\gamma_{\min}.$$

It is less elementary how this shape measure generalizes to tetrahedra or higher-dimensional simplices. For example, neither a tetrahedron's minimum angle within the faces nor its minimum dihedral angle between faces along an edge would be adequate as a shape measure to detect all tetrahedron degeneracies; see Figure 1. The natural generalization of the minimum angle of a triangle is the *minimum solid angle* at the vertices of a simplex.

The classical angle between two linearly independent unit vectors in the plane is the length of the shorter arc on the unit circle connecting them. Solid angles generalize this idea to any dimension. The solid angle of three linearly independent unit vectors in three-dimensional Euclidean space is the surface area of the spherical triangle spanned by those three vectors. In general, the solid angle of n vectors $x_1, \dots, x_n$ in $\mathbb{R}^n$ is the $(n-1)$ -dimensional measure

$$\Omega(x_1, \dots, x_n) := \mathcal{H}^{n-1}(\text{cone}\{x_1, \dots, x_n\} \cap \hat{\mathbb{S}}^{n-1}),$$

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where $\text{cone}\{x_1, \dots, x_n\} := \sum_{i=1}^n \mathbb{R}_0^+ \cdot x_i$ is the convex cone spanned by $x_1, \dots, x_n$ and $\hat{\mathbb{S}}^{n-1}$ denotes the unit sphere of dimension $n-1$. If $x_1, \dots, x_n$ are linearly independent unit vectors, then their solid angle equals the measure of the spherical simplex with vertices $x_1, \dots, x_n$. Given a simplex with vertices $v_0, \dots, v_n \in \mathbb{R}^n$, the solid angle at the vertex v_i is defined as

$$\Omega_i := \Omega(v_0 - v_i, \dots, v_{i-1} - v_i, v_{i+1} - v_i, \dots, v_n - v_i).$$

We examine the minimum solid angle as a simplex shape measure in any dimension. Clearly, the solid angles remain invariant under reflection, rotation, translation, and positive scaling. Well-shaped simplices have large solid angles: the minimum solid angle is bounded from below by fullness of the simplex, up to a dimension-dependent constant. The converse has not been established yet in the literature for higher dimensions; it is easily seen in two dimensions, and Liu and Joe have proven it in three dimensions [19], a considerably more delicate inequality. Generalizing previous work, we show that simplices in any dimension with large solid angles are also well-shaped. We provide polynomial inequalities with explicit dimension-dependent constants and demonstrate that the exponents are optimal.

The main result is the following theorem.

Theorem 1.1. *Let S be an n -dimensional simplex, $n \geq 2$, with diameter δ_S , volume V_S , and minimum solid angle Ω_S . Write Γ for the gamma function. Then*

$$(1.1) \quad C_n^{-1} \Omega_S^2 \leq \mu_S \leq \frac{1}{n} \Omega_S, \quad C_n := n\pi^{\frac{n-1}{2}} \sum_{\ell=1}^{n-2} \binom{n-1}{\ell}^2 \frac{\Gamma(\frac{\ell+1}{2})\Gamma(\frac{n-\ell}{2})}{\Gamma(\frac{n}{2})^2} + \frac{2n\pi^{\frac{n}{2}}}{(n+1)\Gamma(\frac{n}{2})}.$$

In the case $n = 3$, we additionally have

$$(1.2) \quad (12 + 3\pi)^{-1} \Omega_S^2 \leq \mu_S \leq \frac{1}{3} \Omega_S.$$

In dimension three or more, the exponents are optimal.

We briefly review how shape measures are used in different areas of mathematics, engineering, and natural sciences, both in application-oriented and theoretical endeavors.

Some shape measures appear throughout geometric analysis and differential geometry. The *fullness* of an n -dimensional simplex compares its n -dimensional measure against the n -th power of its diameter [28], and the *thickness* of a simplex is the ratio of its smallest height to its diameter [20, 8].

Several shape measures have been proposed throughout the literature of computational geometry and numerical analysis to quantify the quality of simplices [19, 7]. One example is the ratio of the diameter to the radius of the largest inscribed sphere of a simplex, sometimes known as the *chunkiness parameter* (see [5, p. 97, Definition (4.2.16)] and [10, Definition 11.2]). *Algebraic shape measures* are very natural quality indicators for the error analysis of numerical methods: simplices are associated with the affine transformations from regular reference simplices, and their shape quality can therefore be assessed through the properties of the corresponding Jacobian matrices. For example, the spectral condition number of such an affine transformation's Jacobian is a shape measure. While the minima for either face angles or for dihedral angles in a tetrahedron alone are not adequate shape measures, the minima of both face and dihedral angles is a shape measure [2]. Naturally, none of these shape measures depend on the orthogonal coordinate system and the units of distance.

Tetrahedral meshes are widely used in the computational solution of partial differential equations by the finite element method, which is a staple in computational engineering. Ensuring that the tetrahedra are well-shaped is a central problem for the initial mesh generation and for driving adaptive mesh refinement: poorly shaped tetrahedra can adversely affect the discretization errors of the numerical methods, the conditioning of numerical problems, and the accuracy and stability of the numerical solvers. We refer to the literature for overviews on triangular and tetrahedral shape measures [7, 2, 6]. Beyond three dimensions, space-time methods [22, 27] posed on four-dimensional simplicial meshes give us reason to study shape measures in higher dimensions. This manuscript establishes that a family of simplicial meshes in any dimension is shape-regular if and only if the minimum solid angles among all the simplices have a common lower bound.

The notion of solid angle has been known at least since Euclid's *Elements* [21]. The connection to spherical geometry in three or more dimensions has been established over the past centuries [14, 11, 25, 26]. Formulas for solid angles exist in the three-dimensional case, yet no exact elementary expressions in terms of the vector

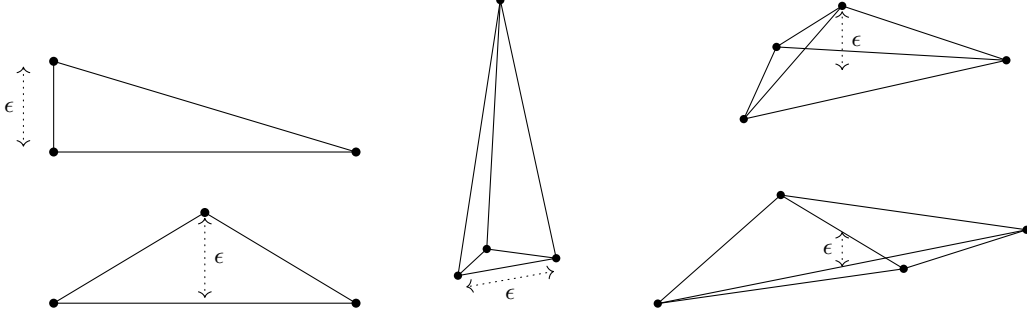


FIGURE 1. Representative examples of simplices degenerating as $\epsilon \rightarrow 0$. Left: triangles degenerate to a line. Middle: a needle tetrahedron degenerating to a single line. Right: tetrahedra degenerating to a triangle (top, pyramid) or a quadrilateral (bottom, sliver). Tetrahedra may degenerate even if dihedral or face angles remain bounded from above and below, but the degeneracy is detected by the minimum solid angle.

coordinates are known in higher dimensions [9, 15, 24]. Solid angles can be expressed in terms of integrals of multivariate Gaussian probability distributions over the positive orthant. Whilst not as commonplace as planar angles, solid angles have remained a staple in several disciplines of engineering and science. The minimum angle is easily seen to be a shape measure for triangles, and the minimum solid angle is a shape measure for tetrahedra [19, 3]. We generalize these observations to arbitrary dimension.

The remainder of this manuscript has the following structure. Section 2 reviews common simplex shape measures and relationships between them. Section 3 defines solid angles and discusses the gnomonic projection. Section 4 shows that the minimum solid angle is a shape measure of simplices in dimension three and higher.

Notation. When the dimension n is understood, $e_1, \dots, e_n \in \mathbb{R}^n$ are the standard orthonormal basis vectors. Write $\mathbf{1}_n$ and $\mathbf{0}_n$ for the n -dimensional vectors whose entries are all equal to 1 or 0, respectively. $\text{Id}_{m \times n}$ is the $m \times n$ matrix whose diagonal entries equal 1 and whose off-diagonal entries are zero. Set $\text{Id}_n := \text{Id}_{n \times n}$.

Given $x \in \mathbb{R}^n$ and $r > 0$, let $\mathbb{B}(x, r)$ be the n -dimensional open ball around x of radius r , and let $\mathbb{S}(x, r)$ be the sphere of dimension $n - 1$ around x of radius r . Write $\hat{\mathbb{B}}^n := \mathbb{B}(\mathbf{0}_n, 1)$ and $\hat{\mathbb{S}}^{n-1} := \mathbb{S}(\mathbf{0}_n, 1)$ for the unit ball and the unit sphere in $\mathbb{R}^n$, respectively. Write $|\hat{\mathbb{B}}^n| := \mathcal{H}^n(\hat{\mathbb{B}}^n)$ and $|\hat{\mathbb{S}}^{n-1}| := \mathcal{H}^{n-1}(\hat{\mathbb{S}}^{n-1})$ for their respective measures. We adopt the convention that the zero-dimensional measure of a point is one, and that the measures are understood in the Hausdorff sense.

Set $\mathbb{R}^+ := (0, \infty)$ and $\mathbb{R}_0^+ := [0, \infty)$. If $a_1, \dots, a_m$ are vectors, then $\text{span}\{a_1, \dots, a_m\}$ is their linear hull, $\text{cone}\{a_1, \dots, a_m\} := \sum_{i=1}^m \mathbb{R}_0^+ \cdot a_i$ denotes their cone (conal hull), and $\text{convex}\{a_1, \dots, a_m\}$ is their convex hull.

2. SIMPLICES AND SHAPE MEASURES

A *simplex* of dimension n is the convex hull of $n + 1$ affinely independent points within Euclidean space. These $n + 1$ points are the *vertices* of the simplex. By definition, no vertex is a convex combination of the others. A simplex is called *regular* if all its edges have the same length. We henceforth assume that $n \geq 1$.

Suppose that S is an n -dimensional simplex with vertices $v_0, \dots, v_n$. We write δ_S for its *diameter*, and we write V_S for its n -dimensional *volume*. The *outer radius* ϱ_S of the simplex S is the minimum radius of any closed ball containing the simplex^a. The *inradius* r_S of the simplex S is the maximum radius of any closed ball (within the affine hull of S) contained in the simplex. The *height* of the i -th vertex is the distance from v_i to the affine hull of $v_0, \dots, v_{i-1}, v_{i+1}, \dots, v_n$. Let ν_S be the smallest among the heights of the $n + 1$ vertices.

We work with several shape measures. The *aspect ratio* α_S , the *thickness* τ_S , the *radius ratio* β_S , and the *fullness* μ_S are

$$\alpha_S := \frac{r_S}{\delta_S}, \quad \tau_S := \frac{\nu_S}{\delta_S}, \quad \beta_S := \frac{r_S}{\varrho_S}, \quad \mu_S := \frac{V_S}{\delta_S^n}.$$

^aWe distinguish the outer radius of S from its *circumradius*, which is the radius of the unique sphere in the affine hull of S that includes all its vertices.

All these measures describe how close a simplex is to being degenerate. To compare these measures, we first recall several elementary geometric identities. For tighter inequalities, some extremal properties of the regular simplex are collected.

Proposition 2.1. *Let S be an n -dimensional simplex.*

- If $\delta_S = 1$, then $\varrho_S \leq C_{J,n} := \sqrt{\frac{n}{2(n+1)}}$ and $V_S \leq C_{\delta,n} := \frac{1}{n!} \sqrt{\frac{n+1}{2^n}}$.
- If $r_S = 1$, then $V_S \geq C_{r,n} := \frac{\sqrt{n^n \sqrt{n+1}^{n+1}}}{n!}$.
- If $\varrho_S = 1$, then $V_S \leq C_{\varrho,n} := \frac{\sqrt{n+1}^{n+1}}{n! \sqrt{n^n}}$.

In each case, equality holds if S is regular.

Proof. Jung's inequality [16] states that n -simplices with unit diameter have outer radius at most $C_{J,n}$. The other inequalities are known extremal properties of the regular simplex; see [18, Remark 1.2], [6, Lemma 11], [4, p. 438], [1, Theorem 2], [13, Theorem 2.2]. $\square$

It is clear that $\varrho_S \leq \delta_S \leq 2\varrho_S$, but the comparison of the outer radius and the diameter can be improved via Jung's theorem:

$$(2.1) \quad C_{J,n}^{-1} \varrho_S \leq \delta_S \leq 2\varrho_S.$$

Let $\nu_0, \dots, \nu_n$ denote the heights of the vertices. Let $A_0, \dots, A_n$ be the measures of the codimension-one faces opposite the respective vertices. Recall

$$(2.2) \quad V_S = \frac{A_i \nu_i}{n}.$$

The largest ball contained within S touches the codimension-one faces tangentially. Its center is called the *incenter* and is equidistant to the affine hulls of the codimension-one faces. Applying (2.2) to each simplex in the simplicial subdivision centered at the incenter, we find

$$V_S = \sum_{i=0}^n \frac{r_S A_i}{n} = r_S \sum_{i=0}^n \frac{A_i}{n} = r_S \sum_{i=0}^n \frac{V_S}{\nu_i}.$$

This shows that r_S equals the harmonic mean of the $n+1$ vertex heights divided by $n+1$:

$$(2.3) \quad r_S = \left(\sum_{i=0}^n \frac{1}{\nu_i} \right)^{-1}.$$

In particular, $r_S \leq \nu_S$. However, the reciprocal heights ν_i^{-1} are the norms of the gradients of the barycentric coordinates of the simplex, and the sum of those gradients equals zero. Therefore, ν_S^{-1} is at most the sum of the other reciprocal heights. Therefore,

$$\nu_S^{-1} \leq r_S^{-1} - \nu_S^{-1}.$$

We conclude that

$$(2.4) \quad \frac{1}{n+1} \nu_S \leq r_S \leq \frac{1}{2} \nu_S.$$

This naturally leads to our first set of comparisons between shape measures:

Proposition 2.2. *We have*

$$\frac{1}{2} \tau_S \geq \alpha_S \geq (n+1)^{-1} \tau_S, \quad C_{J,n} \beta_S \geq \alpha_S \geq \frac{1}{2} \beta_S, \quad \tau_S \geq \beta_S \geq \frac{C_{J,n}^{-1}}{n+1} \tau_S.$$

It is clear that V_S is bounded from below by the volume of its inscribed ball with radius r_S and from above by the volume of the enclosing ball with radius ϱ_S . Proposition 2.1 gives the tighter bounds

$$(2.5) \quad C_{r,n} r_S^n \leq V_S \leq C_{\varrho,n} \varrho_S^n.$$

Using (2.3) and Proposition 2.1 on the codimension-one faces, we obtain the inequality

$$V_S = r_S \sum_{i=0}^n \frac{A_i}{n} \leq \frac{n+1}{n} C_{\delta, n-1} r_S \delta_S^{n-1}.$$

The height of any vertex is bounded by the same vertex's height in any proper subsimplex containing it. Therefore, using (2.2) recursively leads to

$$\frac{1}{n!} \nu_S^n \leq V_S \leq \frac{C_{\delta, n-1}}{n} \nu_S \delta_S^{n-1}.$$

We state our second set of shape measure comparisons.

Proposition 2.3. *We have*

$$\frac{n+1}{n} C_{\delta, n-1} \alpha_S \geq \mu_S \geq C_{r, n} \alpha_S^n, \quad \frac{C_{\delta, n-1}}{n} \tau_S \geq \mu_S \geq \frac{1}{n!} \tau_S^n, \quad \frac{n+1}{n} C_{\delta, n-1} C_{J, n} \beta_S \geq \mu_S \geq \frac{C_{r, n}}{2^n} \beta_S^n.$$

Remark 2.4. Let us illustrate these inequalities in low dimensions. If $n = 2$ and S is a triangle, then the preceding inequalities become

$$\begin{aligned} \frac{1}{2} \tau_S &\geq \alpha_S \geq \frac{1}{3} \tau_S, & \frac{1}{\sqrt{3}} \beta_S &\geq \alpha_S \geq \frac{1}{2} \beta_S, & \tau_S &\geq \beta_S \geq \frac{1}{\sqrt{3}} \tau_S, \\ \frac{3}{2} \alpha_S &\geq \mu_S \geq 3\sqrt{3} \alpha_S^2, & \frac{1}{2} \tau_S &\geq \mu_S \geq \frac{1}{2} \tau_S^2, & \frac{\sqrt{3}}{2} \beta_S &\geq \mu_S \geq \frac{3\sqrt{3}}{4} \beta_S^2. \end{aligned}$$

If $n = 3$ and S is a tetrahedron, then the preceding inequalities become

$$\begin{aligned} \frac{1}{2} \tau_S &\geq \alpha_S \geq \frac{1}{4} \tau_S, & \sqrt{\frac{3}{8}} \beta_S &\geq \alpha_S \geq \frac{1}{2} \beta_S, & \tau_S &\geq \beta_S \geq \frac{1}{\sqrt{6}} \tau_S, \\ \frac{1}{\sqrt{3}} \alpha_S &\geq \mu_S \geq 8\sqrt{3} \alpha_S^3, & \frac{\sqrt{3}}{12} \tau_S &\geq \mu_S \geq \frac{1}{6} \tau_S^3, & \frac{1}{2\sqrt{2}} \beta_S &\geq \mu_S \geq \sqrt{3} \beta_S^3. \end{aligned}$$

3. SOLID ANGLES AS SHAPE MEASURES

Solid angles are higher-dimensional generalizations of the classical angle between two vectors. We discuss solid angles of simplices and how they relate to the fullness. The main result of this section can be paraphrased as follows: *simplices are well-shaped if and only if they do not have small solid angles.*

If $\hat{u}_0, \dots, \hat{u}_m \in \hat{\mathbb{S}}^{n-1}$ are linearly independent, then

$$\mathcal{A}(\hat{u}_0, \dots, \hat{u}_m) := \hat{\mathbb{S}}^{n-1} \cap \text{cone}\{\hat{u}_0, \dots, \hat{u}_m\}$$

is the spherical m -dimensional simplex with vertices $\hat{u}_0, \dots, \hat{u}_m$. The spherical simplex, the cone, and the vertices uniquely identify each other.

Assume that $S \subset \mathbb{R}^n$ is an n -dimensional simplex with vertices $v_0, \dots, v_n$. The *solid angle* of S at vertex v_i is defined as the measure

$$(3.1) \quad \Omega_i(S) := \mathcal{H}^{n-1} \left(\hat{\mathbb{S}}^{n-1} \cap \text{cone}\{v_0 - v_i, \dots, v_{i-1} - v_i, v_{i+1} - v_i, \dots, v_n - v_i\} \right).$$

In other words, the edges adjacent to the vertex v_i define a spherical simplex,

$$\mathcal{A}_i := \mathcal{A} \left(\frac{v_0 - v_i}{\|v_0 - v_i\|}, \dots, \frac{v_{i-1} - v_i}{\|v_{i-1} - v_i\|}, \frac{v_{i+1} - v_i}{\|v_{i+1} - v_i\|}, \dots, \frac{v_n - v_i}{\|v_n - v_i\|} \right),$$

and the solid angle is the measure of that spherical simplex. Alternatively,

$$(3.2) \quad \Omega_i(S) := \lim_{\epsilon \rightarrow 0} \frac{\mathcal{H}^{n-1}(\mathbb{S}(v_i, \epsilon) \cap S)}{\epsilon^{n-1}}.$$

It is easy to see that this quotient attains a constant value for $\epsilon > 0$ small enough.

A distinct formulation of the solid angle occasionally is of great utility. Suppose that the matrix $M_i \in \mathbb{R}^{n \times n}$ has as columns the vectors

$$v_0 - v_i, \quad \dots \quad v_{i-1} - v_i, \quad v_{i+1} - v_i, \quad \dots \quad v_n - v_i,$$

The solid angle at v_i satisfies the representation

$$(3.3) \quad \Omega_i(S) = \pi^{-\frac{n}{2}} \int_{M_i \cdot \mathbb{R}_+^n} \exp(-\|x\|^2) dx = \frac{\det(M_i)}{\pi^{\frac{n}{2}}} \int_{\mathbb{R}_+^n} \exp(-\|M_i x\|^2) dx.$$

as an integral over the positive orthant.

It is a classical result that the sum of the angles within a triangle equals π . The generalization to higher dimension takes a weaker form. The sum of the solid angles in a tetrahedron is at most 2π and can be arbitrarily close to zero [12]. More generally, the sum of the solid angles in an n -dimensional simplex is at most $|\hat{\mathbb{S}}^{n-1}|/2$. This extension to higher dimensions can be embedded in the discussion of convex polytopes [23, (24)] and random polytopes [17, Theorem 3.10] but also admits an elementary proof that is inspired by the classical planar case. We include it for completeness.

Theorem 3.1. *Let S be an n -dimensional simplex with vertices $v_0, \dots, v_n \in \mathbb{R}^n$, for $n \geq 2$. Then*

$$(3.4) \quad \sum_{i=0}^n \Omega_i(S) \leq \frac{|\hat{\mathbb{S}}^{n-1}|}{2}.$$

In particular, the minimum solid angle of S is at most $|\hat{\mathbb{S}}^{n-1}|/(2n+2)$.

Proof. Without loss of generality, after a translation, $v_0 = 0$. Suppose that $x \in \mathbb{R}^n$. There exist unique $\lambda_1, \dots, \lambda_n, \mu_0, \dots, \mu_{n-1} \in \mathbb{R}$ such that

$$x = \sum_{i=1}^n \lambda_i (v_i - v_0) = \sum_{i=0}^{n-1} \mu_i (v_i - v_n).$$

We rearrange this equation,

$$(-\sum_{i=1}^n \lambda_i) v_0 + \sum_{i=1}^n \lambda_i v_i + \lambda_n v_n = \mu_0 v_0 + \sum_{i=1}^{n-1} \mu_i v_i + (-\sum_{i=0}^{n-1} \mu_i) v_n,$$

and notice that $\lambda_i = \mu_i$ for $1 \leq i \leq n-1$. In addition, $\lambda_n = -\sum_{i=0}^{n-1} \mu_i$. Consider the open cones

$$C_0^+ := \sum_{i=1}^n \mathbb{R}^+ \cdot (v_i - v_0), \quad C_0^- := -C_0^+, \quad C_n^+ := \sum_{i=0}^{n-1} \mathbb{R}^+ \cdot (v_i - v_n), \quad C_n^- := -C_n^+.$$

Clearly, $C_0^+ \cap C_0^- = C_n^+ \cap C_n^- = \emptyset$. If $x \in C_0^+$, then positivity of $\lambda_1, \dots, \lambda_{n-1}$ shows that $x \notin C_n^-$, and positivity of λ_n shows that $x \notin C_n^+$. Likewise, C_n^+ is disjoint from C_0^- . We conclude that the spherical simplices defined by the $n+1$ vertices, and their reflections across the origin,

$$\mathcal{A}_0, \mathcal{A}_1, \dots, \mathcal{A}_n, -\mathcal{A}_0, -\mathcal{A}_1, \dots, -\mathcal{A}_n,$$

are essentially disjoint within the unit sphere of dimension $n-1$. The inequality (3.4) is now evident. $\square$

The minimum solid angle

$$\Omega_S := \min\{\Omega_0(S), \Omega_1(S), \dots, \Omega_n(S)\}$$

of the simplex S is a candidate for a shape measure. The simplex S is contained in the volume suspended by any vertex and its opposite spherical simplex, scaled by the radius δ_S . Hence,

$$V_S \leq \frac{\Omega_S}{|\hat{\mathbb{S}}^{n-1}|} |\hat{\mathbb{B}}^n| \delta_S^n = \frac{\delta_S^n}{n} \Omega_S.$$

Therefore,

$$\mu_S \leq \frac{1}{n} \Omega_S.$$

This means that well-shaped simplices do not have small solid angles. The purpose of this manuscript is to establish the converse implication.

Example 3.2. Let $\epsilon > 0$. Consider a simplex S_ϵ with vertices

$$v_0 = 0, \quad v_1 = e_1, \quad v_2 = e_1 + \epsilon e_2, \quad \dots \quad v_n = e_1 + \epsilon e_n.$$

Obviously, $\delta_{S_\epsilon} = \sqrt{1 + \epsilon^2}$ and $V_{S_\epsilon} = \epsilon^{n-1}/n!$, and therefore

$$\mu_{S_\epsilon} = V_{S_\epsilon} \delta_{S_\epsilon}^{-n} = \frac{\epsilon^{n-1}}{n!(1 + \epsilon^2)^{\frac{n}{2}}}.$$

Always, $\Omega_1 = \hat{\mathbb{S}}^{n-1}/4$, and the solid angles at $v_2, \dots, v_n$ monotonically increase towards a constant value as ϵ decreases. For the solid angle at v_0 , consider the matrix

$$M_0 := \begin{pmatrix} 1 & 1 & \dots & 1 \\ & \epsilon & & \\ & & \ddots & \\ & & & \epsilon \end{pmatrix}.$$

If $x \in \mathbb{R}^n$ with positive coordinates, then one easily verifies

$$\|M_0 x\|^2 \leq (n + \epsilon^2)\|x\|^2, \quad \|M_0 x\|^2 \geq \|x\|^2.$$

Applying the lower and upper bound in the representation (3.3) of the solid angles as orthant integral and performing a substitution, the following inequalities emerge:

$$\frac{\epsilon^{n-1}}{n!(n+\epsilon)^{\frac{n}{2}}} \leq \Omega_0 \leq \frac{\epsilon^{n-1}}{n!2^n}.$$

This shows that any upper bound for the fullness in terms of the minimum solid angle, in two or more dimensions, must be at least linear.

Let $\mathbb{H}_n$ be the northern halfspace, and let $\mathbb{E}_{n-1} \subset \mathbb{R}^n$ be the tangent plane at the north pole,

$$\mathbb{H}_n := \{(x_1, \dots, x_n) \in \mathbb{R}^n \mid x_n > 0\}, \quad \mathbb{E}_{n-1} := \{x \in \mathbb{R}^n \mid x_n = 1\}.$$

The gnomonic projection is

$$\Phi : \hat{\mathbb{S}}^{n-1} \cap \mathbb{H}_n \rightarrow \mathbb{E}_{n-1}, \quad (x_1, \dots, x_{n-1}, x_n) \mapsto (x_1/x_n, \dots, x_{n-1}/x_n, 1).$$

Its inverse takes the form

$$\Phi^{-1} : \mathbb{E}_{n-1} \rightarrow \hat{\mathbb{S}}^{n-1} \cap \mathbb{H}_n, \quad y \mapsto \frac{y}{\|y\|}.$$

The gnomonic projection Φ maps spherical simplices in the northern hemisphere $\hat{\mathbb{S}}^{n-1} \cap \mathbb{H}_n$ to simplices within the tangential plane $\mathbb{E}_{n-1}$.

Lemma 3.3. *Let $\hat{u}_0, \dots, \hat{u}_m \in \hat{\mathbb{S}}^{n-1} \cap \mathbb{H}_n$ be linearly independent unit vectors in the northern hemisphere. Write $u_i := \Phi(\hat{u}_i)$ for $0 \leq i \leq m$. Then $\text{convex}\{u_0, \dots, u_m\}$ is an m -dimensional simplex with vertices $u_0, \dots, u_m$ and*

$$\Phi(\mathcal{A}(\hat{u}_0, \dots, \hat{u}_m)) = \text{convex}\{u_0, \dots, u_m\}.$$

Proof. Write $C := \text{cone}\{\hat{u}_0, \dots, \hat{u}_m\}$ and $\mathcal{A} := \mathcal{A}(\hat{u}_0, \dots, \hat{u}_m)$. We know that $K := \mathbb{E}_{n-1} \cap C$ is a simplex with vertices $u_0, \dots, u_m$. By definition, $\mathcal{A} = \hat{\mathbb{S}}^{n-1} \cap C$. Since Φ preserves half-lines starting at the origin, $\Phi(\mathcal{A}) = K$ is evident. $\square$

Lemma 3.4. *For all $y \in \mathbb{E}_{n-1}$ we have*

$$\det \left(D\Phi|_y^{-t} D\Phi|_y^{-1} \right) = \|y\|^{-2n}.$$

Proof. Recall that the normalization mapping $\tilde{\Phi} : \mathbb{R}^n \setminus \{0\} \rightarrow \hat{\mathbb{S}}^{n-1}$ from the punctured Euclidean space onto the unit sphere, which is defined by $\tilde{\Phi}(y) := y/\|y\|$, has a Jacobian of the form

$$D\tilde{\Phi}[y] = \frac{1}{\|y\|} \text{Id}_{n \times n} - \frac{y \otimes y^t}{\|y\|^3}.$$

The Jacobian of Φ^{-1} at any $y \in \mathbb{E}_{n-1}$ in tangential direction $v \in \mathbb{R}^n$, where we write $y = (y_1, \dots, y_{n-1}, 1)$ and $v = (v_1, \dots, v_{n-1}, 0)$ and abbreviate $z := (y_1, \dots, y_{n-1})$, takes the following form:

$$D\Phi^{-1}[y] = (\|z\|^2 + 1)^{-\frac{1}{2}} \text{Id}_{n \times (n-1)} - (\|z\|^2 + 1)^{-\frac{3}{2}} ((z, 0) + e_n) \otimes z^t.$$

We abbreviate $s = \|z\|^2 + 1$ and find

$$(D\Phi^{-1}[y])^t (D\Phi^{-1}[y]) = \frac{1}{s} \text{Id}_{n-1} - \frac{2}{s^2} z \otimes z^t + \frac{\|(z, 1)\|^2}{s^3} z \otimes z^t = \frac{1}{s} \left(\text{Id}_{n-1} - \frac{z \otimes z^t}{s} \right).$$

Therefore, the matrix determinant lemma applies and gives

$$\det \left(\text{Id}_{n-1} - \frac{z \otimes z^t}{s} \right) = 1 - \frac{z^t z}{s} = 1 - \frac{\|z\|^2}{\|z\|^2 + 1} = \frac{1}{s}.$$

The desired determinant formula follows. $\square$

Remark 3.5. The gnomonic projection of the one-dimensional northern hemisphere, the upper semicircle, elegantly relates to trigonometric identities. Let $x = (x_1, x_2) \in \hat{\mathbb{S}}^1 \cap \mathbb{H}_2$ be in the northern hemisphere, and let $(z, 1) = \Phi(x)$. Let $\alpha = \arctan(z) \in (-\pi/2, \pi/2)$. Then $x_1 = \sin(\alpha)$ and $x_2 = \cos(\alpha)$. At the same time, $x_1 = \sin(\arctan(z)) = z/\sqrt{1+z^2}$ and $x_2 = \cos(\arctan(z)) = 1/\sqrt{1+z^2}$. From $z = x_1/x_2$ and $1 = x_1^2 + x_2^2$, converse identities $z = \tan(\arcsin(x_1)) = x_1/\sqrt{1-x_1^2}$ and $|z| = \tan(\arccos(x_2)) = \sqrt{1-x_2^2}/x_2$ are derived.

The area of spherical simplices, and thus solid angles, can be formulated as a Euclidean integral. Suppose that $\hat{u}_1, \dots, \hat{u}_n \in \hat{\mathbb{S}}^{n-1} \cap \mathbb{H}_n$ are linearly independent unit vectors in the northern hemisphere. Then

$$\mathcal{A} := \hat{\mathbb{S}}^{n-1} \cap \text{cone}\{\hat{u}_1, \dots, \hat{u}_n\}$$

is a spherical simplex that is mapped along the gnomonic projection to the simplex

$$K := \Phi(\mathcal{A}) = \text{convex}\{\Phi(\hat{u}_1), \dots, \Phi(\hat{u}_n)\}.$$

The area of the spherical simplex admits the representation as a weighted Euclidean integral

$$\mathcal{H}^{n-1}(\mathcal{A}) = \int_K \|y\|^{-n} d\mathcal{H}^{n-1}.$$

Notice that the integrand is non-singular and smooth because $K \subset \mathbb{E}_{n-1}$. This reformulation of solid angles is central to our analysis of solid angles.

4. THE MINIMUM SOLID ANGLE

We first address the special case of a tetrahedron, $n = 3$, and then the case of simplices in any dimension n .

Theorem 4.1. *Let $S \subset \mathbb{R}^3$ be a tetrahedron. We have the estimate*

$$(4.1) \quad C_{\text{tet}}^{-1} \Omega_S^2 \leq \mu_S \leq \frac{1}{3} \Omega_S, \quad C_{\text{tet}} := 12 + 3\pi.$$

Proof. Let S be a tetrahedron with vertices $v_0, v_1, v_2, v_3 \in \mathbb{R}^3$. It suffices to consider the case $\delta_S = 1$ because the rest follows by a scaling argument. Without loss of generality, after rotation, translation, and relabeling, $v_0 = 0$ and $v_3 = e_3$. Since the edge from v_0 to v_3 is the diameter of S , the remaining vertices have the form

$$v_1 = (f_1, \theta_1), \quad v_2 = (f_2, \theta_2),$$

where $f_1, f_2 \in \mathbb{R}^2$ and $\theta_1, \theta_2 \in (0, 1)$, and f_1, f_2 are linearly independent. These satisfy the inequalities

$$\|f_1\|^2 + \theta_1^2 \leq 1, \quad \|f_2\|^2 + \theta_2^2 \leq 1, \quad \|f_1\|^2 + (1 - \theta_1)^2 \leq 1, \quad \|f_2\|^2 + (1 - \theta_2)^2 \leq 1.$$

The vertices v_1, v_2, v_3 lie in the northern half-space $\mathbb{H}_3$. Let $\hat{v}_1, \hat{v}_2$ be the unit vectors in the directions of v_1, v_2 , respectively. Their gnomonic projections onto the northern tangential plane $\mathbb{E}_2$ are

$$\Phi(\hat{v}_1) = (f_1/\theta_1, 1), \quad \Phi(\hat{v}_2) = (f_2/\theta_2, 1).$$

Taking the gnomonic projection of the spherical triangle opposite to v_0 onto $\mathbb{E}_2$ and omitting the third coordinate gives the triangle

$$K_0 = \text{convex} \left\{ 0, \frac{f_1}{\theta_1}, \frac{f_2}{\theta_2} \right\} \subset \mathbb{R}^2.$$

Therefore, the solid angle $\Omega_0 := \Omega_0(S)$ equals the planar integral

$$\Omega_0 = \iint_{K_0} (1 + \|z\|^2)^{-\frac{3}{2}} dz.$$

A fully analogous argument rewrites the solid angle $\Omega_3 := \Omega_3(S)$ as the planar integral

$$\Omega_3 = \iint_{K_3} (1 + \|z\|^2)^{-\frac{3}{2}} dz, \quad K_3 = \text{convex} \left\{ 0, \frac{f_1}{1 - \theta_1}, \frac{f_2}{1 - \theta_2} \right\}.$$

We define the new triangle $F = \text{convex}\{0, f_1, f_2\}$. Let $\alpha \in (0, \pi)$ be the angle between f_1 and f_2 . By Cavalieri's principle, $3V_S = V_F$. By construction, the measures satisfy

$$V_{K_0} = \theta_1^{-1} \theta_2^{-1} V_F, \quad V_{K_3} = (1 - \theta_1)^{-1} (1 - \theta_2)^{-1} V_F, \quad 3V_S = V_F = \frac{1}{2} \sin(\alpha) b_1 b_2, \\ b_1 := \|f_1\|, \quad b_2 := \|f_2\|.$$

We can now estimate the solid angles. First, we notice the simple estimates

$$\Omega_0 \leq V_{K_0}, \quad \Omega_3 \leq V_{K_3}.$$

Since K_0 and K_3 are triangles with the origin as a vertex, they lie in a radial sector of $\mathbb{R}^2$ with angle α . Thus,

$$\Omega_0 \leq \frac{\alpha}{2\pi} \iint_{\mathbb{R}^2} (1 + \|z\|^2)^{-\frac{3}{2}} dz = \alpha \int_0^\infty (1 + r^2)^{-\frac{3}{2}} r dr = \alpha \int_0^\infty \left(-(1 + r^2)^{-\frac{1}{2}} \right)' dr = \alpha.$$

The same argument applies to K_3 . Thus, $\Omega_0 \leq \alpha$, and $\Omega_3 \leq \alpha$, which yields the product estimates

$$\Omega_S^2 \leq \Omega_0 \Omega_3 \leq \theta_1^{-1} \theta_2^{-1} V_F \alpha, \quad \Omega_S^2 \leq \Omega_0 \Omega_3 \leq (1 - \theta_1)^{-1} (1 - \theta_2)^{-1} V_F \alpha.$$

At the same time, Theorem 3.1 states that the minimum solid angle of a tetrahedron is at most $\pi/2$, hence

$$\Omega_S^2 \leq \frac{\pi}{2} \Omega_0 \leq \theta_1^{-1} \theta_2^{-1} \frac{\pi}{2} V_F, \quad \Omega_S^2 \leq \frac{\pi}{2} \Omega_3 \leq (1 - \theta_1)^{-1} (1 - \theta_2)^{-1} \frac{\pi}{2} V_F.$$

The inclusion $K_0 \subseteq [0, 1] \theta_1^{-1} f_1 + \mathbb{R}_0^+ \theta_2^{-1} f_2$ gives, after an orthogonal transformation that aligns f_2 with the second coordinate direction,

$$\Omega_0 \leq I(\theta_1^{-1} b_1, \alpha), \quad I(\theta_1^{-1} b_1, \alpha) := \int_0^{\theta_1^{-1} b_1 \sin(\alpha)} \int_{x \cot(\alpha)}^\infty (1 + x^2 + y^2)^{-\frac{3}{2}} dy dx.$$

Abbreviate $B = \cot(\alpha)x$. We substitute $y = \sqrt{1 + x^2} \tan(\varphi)$ and find

$$\begin{aligned} \int_B^\infty (1 + x^2 + y^2)^{-\frac{3}{2}} dy &= \int_{\arctan(B/\sqrt{1+x^2})}^{\frac{\pi}{2}} \frac{\sec(\varphi)^2}{(1 + x^2)(1 + \tan(\varphi)^2)^{\frac{3}{2}}} d\varphi = \frac{1}{1 + x^2} \int_{\arctan(B/\sqrt{1+x^2})}^{\frac{\pi}{2}} \frac{\sec(\varphi)^2}{\sec(\varphi)^3} d\varphi \\ &= \frac{1}{1 + x^2} \int_{\arctan(B/\sqrt{1+x^2})}^{\frac{\pi}{2}} \cos(\varphi) d\varphi = \frac{1 - \sin(\arctan(B/\sqrt{1+x^2}))}{1 + x^2} \\ &= \frac{1}{1 + x^2} - \frac{\sin \arctan(\cot(\alpha) \frac{x}{\sqrt{1+x^2}})}{1 + x^2}. \end{aligned}$$

As in Remark 3.5, one can simplify

$$\sin \arctan(\cot(\alpha) \frac{x}{\sqrt{1+x^2}}) = \frac{\cot(\alpha)x}{\sqrt{1 + (1 + \cot^2(\alpha))x^2}} = (\sin^2(\alpha) + x^2)^{-\frac{1}{2}} \cos(\alpha)x.$$

We bound the integral of this expression in x from 0 to $\theta_1^{-1} b_1 \sin(\alpha)$. For any $D > 0$, we notice

$$\begin{aligned} \partial_D I(D, \alpha) &:= \partial_D \int_0^{D \sin(\alpha)} \frac{1}{1 + x^2} - \frac{\cos(\alpha)x}{(1 + x^2)(\sin^2(\alpha) + x^2)^{\frac{1}{2}}} dx \\ &= \frac{\sin(\alpha)}{1 + D^2 \sin(\alpha)^2} - \frac{\cos(\alpha) D \sin(\alpha)^2}{(1 + D^2 \sin(\alpha)^2)(\sin^2(\alpha) + D^2 \sin(\alpha)^2)^{\frac{1}{2}}} \\ &= \frac{\sin(\alpha)}{1 + D^2 \sin(\alpha)^2} - \frac{\cos(\alpha) \sin(\alpha) D}{(1 + D^2 \sin(\alpha)^2)(1 + D^2)^{\frac{1}{2}}} \\ &= \frac{\sin(\alpha)}{1 + D^2 \sin(\alpha)^2} (1 - \cos(\alpha) \sin(\arctan(D))). \end{aligned}$$

The integral $I(\theta_1^{-1} b_1, \alpha)$ is increasing in α up to $\pi/2$. If $\alpha = \pi/2$, then the bound $\partial_D I(D, \alpha) \leq 1$ leads to $I(\theta_1^{-1} b_1, \alpha) \leq \theta_1^{-1} b_1$, so we consider now the case where $0 < \alpha < \pi/2$ or $\pi/2 < \alpha < \pi$. We write

$\varphi := \sin(\arctan(D)) \in (0, 1)$, so that $D = \varphi/\sqrt{1-\varphi^2}$. One observes

$$\begin{aligned} & \frac{\sin(\alpha)}{1 + D^2 \sin(\alpha)^2} (1 - \cos(\alpha) \sin(\arctan(D))) \\ &= \frac{\sin(\alpha)(1 - \varphi^2)(1 - \cos(\alpha)\varphi)}{1 - \varphi^2 + (1 - \varphi^2)D^2 \sin(\alpha)^2} \\ &= \frac{\sin(\alpha)(1 - \varphi^2)(1 - \cos(\alpha)\varphi)}{1 - \varphi^2 + \sin(\alpha)^2 \varphi^2} = \frac{\sin(\alpha)(1 - \varphi^2)(1 - \cos(\alpha)\varphi)}{1 - \cos(\alpha)^2 \varphi^2} = \frac{\sin(\alpha)(1 - \varphi^2)}{1 + \cos(\alpha)\varphi} = \tan(\alpha) \frac{1 - \varphi^2}{\sec(\alpha) + \varphi}. \end{aligned}$$

The last quotient as a function in φ vanishes at -1 and $+1$ and is positive over the interval $(-1, 1)$. Therefore, we seek its critical points at the solutions of the quadratic equation

$$\varphi^2 + 2\varphi \sec(\alpha) + 1 = 0.$$

Its two roots are $\varphi = -\sec(\alpha) \pm \sqrt{\sec(\alpha)^2 - 1}$. We only take the critical point within $(-1, 1)$. If $0 < \alpha < \pi/2$, then this is $\varphi = -\sec(\alpha) + \sqrt{\sec^2(\alpha) - 1}$. If $\pi/2 < \alpha < \pi$, then this is $\varphi = -\sec(\alpha) - \sqrt{\sec^2(\alpha) - 1}$. In either case, $\varphi = \tan(\alpha) - \sec(\alpha)$. Therefore,

$$\begin{aligned} \tan(\alpha) \frac{1 - \varphi^2}{\sec(\alpha) + \varphi} &= 1 - (\sec(\alpha) - \tan(\alpha))^2 \\ &= 1 - \left(\frac{1 - \sin(\alpha)}{\cos(\alpha)} \right)^2 = \frac{2 \sin(\alpha) - 2 \sin(\alpha)^2}{\cos(\alpha)^2} = 2 \sin(\alpha) \frac{1 - \sin(\alpha)}{1 - \sin(\alpha)^2} = \frac{2 \sin(\alpha)}{1 + \sin(\alpha)}. \end{aligned}$$

We integrate $\partial_D I(D, \alpha)$ from 0 to b_1/θ_1 , thus bounding $I(b_1/\theta_1, \alpha)$. For all $0 < \alpha < \pi$, we have verified

$$I(\theta_1^{-1} b_1, \alpha) \leq \theta_1^{-1} b_1 \frac{2 \sin(\alpha)}{1 + \sin(\alpha)}.$$

This leads to the following upper estimate for Ω_0 , another upper estimate based on the second vertex, and analogous upper bounds for the opposite solid angle Ω_3 :

$$\Omega_0 \leq \frac{2b_1\theta_1^{-1} \sin(\alpha)}{1 + \sin(\alpha)}, \quad \Omega_0 \leq \frac{2b_2\theta_2^{-1} \sin(\alpha)}{1 + \sin(\alpha)}, \quad \Omega_3 \leq \frac{2b_1(1 - \theta_1)^{-1} \sin(\alpha)}{1 + \sin(\alpha)}, \quad \Omega_3 \leq \frac{2b_2(1 - \theta_2)^{-1} \sin(\alpha)}{1 + \sin(\alpha)}.$$

Along these lines of reasoning, the following product estimates are established:

$$\Omega_0 \Omega_3 \leq \theta_1^{-1} (1 - \theta_2)^{-1} b_1 b_2 \sin(\alpha) \frac{4 \sin(\alpha)}{(1 + \sin(\alpha))^2} \leq \theta_1^{-1} (1 - \theta_2)^{-1} 2V_F, \quad \Omega_0 \Omega_3 \leq (1 - \theta_1)^{-1} \theta_2^{-1} 2V_F.$$

Here, we have used that $4 \sin(\alpha) \leq (1 + \sin(\alpha))^2$ because $2\sqrt{\sin(\alpha)} \leq 1 + \sin(\alpha)$. This leads to the following sum:

$$\Omega_S^2 = \Omega_S^2 (\theta_1 \theta_2 + \theta_1 (1 - \theta_2) + (1 - \theta_1) \theta_2 + (1 - \theta_1)(1 - \theta_2)) \leq 2 \left(\frac{\pi}{2} + 2 \right) V_F = (12 + 3\pi) V_S.$$

Obviously, this proves (4.1) in the case $\delta_S = 1$. Finally, the assumption $\delta_S = 1$ can be omitted by a scaling argument. This proves the desired estimate. $\square$

Remark 4.2. We remark on the geometric intuition in this proof. First, V_F is proportional to V_S by Cavalieri's principle. If v_1 and v_2 are closer to v_3 , say, $\theta_1, \theta_2 > \frac{1}{2}$, then the solid angle $\Omega_0 \leq V_{K_0} \leq 4V_F$, since the area distortion of the gnomonic projection is bounded in this regime. The solid angle Ω_3 is at most 2π by Theorem 3.1. If instead v_1 and v_2 are closer to v_0 , say, $\theta_1, \theta_2 \leq \frac{1}{2}$, we reverse the roles of which solid angle is controlled and which is estimated only uniformly. In either case, the product $\Omega_0 \Omega_3$ is controlled.

It remains to address the mixed case. If $\theta_1 > \frac{1}{2}$ but $\theta_2 \leq \frac{1}{2}$, then K_0 is contained within the strip $[0, \theta_1^{-1} \|f_1\| \sin(\alpha)] \times \mathbb{R}$, after an orthogonal change of coordinates. The solid angle Ω_0 is bounded by a fixed multiple of the width $\theta_1^{-1} \|f_1\| \sin(\alpha) \leq 2 \|f_1\| \sin(\alpha)$. An analogous estimate applies to the opposite solid angle Ω_3 . Each of these solid angles is controlled by one leg of the triangle F , and hence their product is controlled by both legs. This allows us to use $2V_F = \|f_1\| \|f_2\| \sin(\alpha)$.

Hence, after picking in accordance to whether each of v_1 and v_2 is closer to v_0 or v_3 , we conduct a case distinction and bound $\Omega_0 \Omega_3$ by a multiple of V_S . The proof of Theorem 4.1 refines this idea: instead of picking the worst possible bound, we take a weighted average of these estimates.

Example 4.3. We show that the quadratic exponent in Theorem 4.1 is optimal and cannot be lowered. Suppose that $\epsilon > 0$. Inspired by the sliver example in Figure 1, the tetrahedron S with vertices

$$v_0 = -\epsilon e_1, \quad v_1 = \epsilon e_1, \quad v_2 = e_2 - \epsilon e_3, \quad e_2 = e_2 + \epsilon e_3,$$

is studied for $\epsilon > 0$ small. $\delta_S = \sqrt{1 + 2\epsilon^2}$, and a simple determinant calculation gives $V_S = 2\epsilon^2/3$. Write $d_{ij} := \|v_i - v_j\|$ for $0 \leq i < j \leq 3$. Then $d_{01} = d_{23} = 2\epsilon$ and $d_{02} = d_{03} = d_{12} = d_{13} = \sqrt{1 + 2\epsilon^2}$. The solid angles of S are identical because of symmetry. We use a theorem by Liu and Joe [19, Theorem 2.2],

$$\sin(\Omega_0(S)/2) = 12V_S \cdot \prod_{1 \leq i < j \leq 3} (d_{0i} + d_{0j} + d_{ij})^{-\frac{1}{2}} (d_{0i} + d_{0j} - d_{ij})^{-\frac{1}{2}}.$$

A direct calculation gives

$$\sin(\Omega_0(S)/2) = \epsilon(1 + 3\epsilon^2 + 2\epsilon\sqrt{1 + 2\epsilon^2})^{-1}.$$

Write $k(\epsilon) := 1 + 3\epsilon^2 + 2\epsilon\sqrt{1 + 2\epsilon^2}$. Then

$$\Omega_0(S)/2 = \epsilon k(\epsilon) / \sqrt{1 + \epsilon^2 k(\epsilon)^2} = \epsilon(k(\epsilon)^{-2} + \epsilon^2)^{-\frac{1}{2}}.$$

Clearly, for any $1 \leq p < \infty$ we have

$$\Omega_S^p / V_S = \frac{3}{2} \epsilon^{p-2} 2^p k(\epsilon)^{-\frac{p}{2}}.$$

It is now apparent that $p = 2$ is the lowest possible exponent in the inequality 4.1, and $C_{\text{tet}} \geq 6$.

We analyze the relationship between the fullness and the minimum solid angle of higher dimensional simplices.

Theorem 4.4. *Let $S \subset \mathbb{R}^n$ be an n -dimensional simplex with $n \geq 2$. We have the estimate*

$$(4.2) \quad C_n^{-1} \Omega_S^2 \leq \mu_S \leq \frac{1}{n} \Omega_S, \quad C_n := n\pi^{\frac{n-1}{2}} \sum_{\ell=1}^{n-2} \binom{n-1}{\ell}^2 \frac{\Gamma(\frac{\ell+1}{2})\Gamma(\frac{n-\ell}{2})}{\Gamma(\frac{n}{2})^2} + \frac{2n\pi^{\frac{n}{2}}}{(n+1)\Gamma(\frac{n}{2})}.$$

Proof. The upper bound has already been shown. We address the lower bound. Suppose that S has diameter equal to 1. Without loss of generality, after rotation, translation, and relabeling of vertices,

$$v_0 = 0, \quad v_n = e_n, \quad \|v_0 - v_n\| = 1.$$

Now, the remaining vertices $v_1, \dots, v_{n-1}$ can be written as

$$v_i = (f_i, \theta_i), \quad f_i \in \mathbb{R}^{n-1}, \quad \theta_i \in \mathbb{R}.$$

Since the edge from v_0 to v_n realizes the diameter of S , for all $1 \leq i \leq n-1$:

$$f_i \neq 0, \quad 0 < \theta_i < 1, \quad \theta_i^2 + \|f_i\|^2 \leq 1, \quad (1 - \theta_i)^2 + \|f_i\|^2 \leq 1.$$

We write $\Omega_0 := \Omega_0(S)$ and $\Omega_n := \Omega_n(S)$ as integrals. The vertices $v_1, \dots, v_{n-1}$ are located in the northern half-space $\mathbb{H}_n$. Let $\hat{v}_1, \dots, \hat{v}_{n-1}$ be their radial projections on the northern hemisphere $\mathbb{H}_n \cap \hat{\mathbb{S}}^{n-1}$, so that $\hat{v}_i = v_i / \|v_i\|$. Their gnomonic projections onto the northern tangential plane $\mathbb{E}_{n-1}$ are

$$\Phi(\hat{v}_i) = (f_i / \theta_i, 1) \in \mathbb{E}_{n-1}.$$

We take the gnomonic projection of the spherical simplex at v_0 to the northern tangential plane $\mathbb{E}_{n-1}$. Omitting the last coordinate, the image of the spherical simplex is the lower-dimensional simplex

$$(4.3) \quad K_0 := \text{convex} \left\{ 0, \frac{f_1}{\theta_1}, \dots, \frac{f_{n-1}}{\theta_{n-1}} \right\} \subset \mathbb{R}^{n-1}.$$

Therefore,

$$(4.4) \quad \Omega_0 = \int_{K_0} (1 + \|z\|^2)^{-\frac{n}{2}} dz.$$

A completely analogous argument applies to the solid angle Ω_n at the vertex v_n , with the main change being that factors θ_i are replaced by $1 - \theta_i$. Accordingly,

$$\Omega_n = \int_{K_n} (1 + \|z\|^2)^{-\frac{n}{2}} dz, \quad K_n := \text{convex} \left\{ 0, \frac{f_1}{1 - \theta_1}, \dots, \frac{f_{n-1}}{1 - \theta_{n-1}} \right\}.$$

Next, we consider any partition of the indices $\{1, \dots, n-1\}$ into two sets: let $I_A, I_B \subseteq \{1, \dots, n-1\}$ with

$$(4.5) \quad I_A \cap I_B = \emptyset, \quad I_A \cup I_B = \{1, \dots, n-1\},$$

and define cardinalities

$$\ell_A := |I_A|, \quad \ell_B := |I_B|.$$

Notice that $\ell_A + \ell_B = n-1$. We define the subspaces

$$U_A := \text{span}\{f_i \mid i \in I_A\}, \quad U_B := \text{span}\{f_i \mid i \in I_B\},$$

and the orthogonal projections onto their respective orthogonal complements

$$P_A : \mathbb{R}^{n-1} \rightarrow U_A^\perp, \quad P_B : \mathbb{R}^{n-1} \rightarrow U_B^\perp.$$

Since $V_S > 0$, the vectors $f_1, \dots, f_{n-1}$ are linearly independent. Hence, $\dim U_A = \ell_A$ and $\dim U_B = \ell_B$.

We also define the following simplices and their corresponding volumes:

$$F := \text{convex}\{0, f_1, \dots, f_{n-1}\}, \quad F_A := \text{convex}\{0, f_i \mid i \in I_A\}, \quad F_B := \text{convex}\{0, f_i \mid i \in I_B\},$$

$$V_F := \mathcal{H}^{n-1}(F), \quad V_A := \mathcal{H}^{\ell_A}(F_A), \quad V_B := \mathcal{H}^{\ell_B}(F_B).$$

Due to Cavalieri's principle, S has the same measure as the convex hull of e_n and F (embedded in the coordinate plane $e_n^\perp$). Hence, $n!V_S = (n-1)!V_F$. Again via Cavalieri's principle, F has the same measure as the convex hull of F_A and the projected simplex P_AF_B . Similarly, F has the same measure as the convex hull of F_B and the projected simplex P_BF_A . Consequently,

$$\mathcal{H}^{\ell_A}(F_A)\mathcal{H}^{\ell_B}(P_AF_B) = \frac{(n-1)!}{\ell_A!\ell_B!}V_F, \quad \mathcal{H}^{\ell_A}(P_BF_A)\mathcal{H}^{\ell_B}(F_B) = \frac{(n-1)!}{\ell_A!\ell_B!}V_F.$$

In particular,

$$(4.6) \quad \mathcal{H}^{\ell_B}(P_AF_B) = \frac{(n-1)!}{\ell_A!\ell_B!} \frac{V_F}{V_A}, \quad \mathcal{H}^{\ell_A}(P_BF_A) = \frac{(n-1)!}{\ell_A!\ell_B!} \frac{V_F}{V_B}.$$

Since the projections are orthogonal,

$$(4.7) \quad \mathcal{H}^{\ell_B}(P_AF_B) \leq V_B, \quad \mathcal{H}^{\ell_A}(P_BF_A) \leq V_A, \quad V_F \leq \frac{\ell_A!\ell_B!}{(n-1)!}V_AV_B.$$

This gives rise to estimates for the projected simplices P_BK_0 and P_AK_n . Recall that K_0 is the convex hull of the origin and $f_1/\theta_1, \dots, f_{n-1}/\theta_{n-1}$. Therefore, P_BK_0 is the convex hull of the origin and P_Bf_i/θ_i , where $i \in I_A$. Thus,

$$(4.8) \quad \mathcal{H}^{\ell_A}(P_BK_0) = \prod_{i \in I_A} \theta_i^{-1} \mathcal{H}^{\ell_A}(P_BF_A).$$

Along a completely analogous line of reasoning, one obtains

$$(4.9) \quad \mathcal{H}^{\ell_B}(P_AK_n) = \prod_{i \in I_B} (1 - \theta_i)^{-1} \mathcal{H}^{\ell_B}(P_AF_B).$$

We finally bound the integrals. Obviously, $K_0 \subseteq P_BK_0 + U_B$. The latter is an orthogonal sum. Fubini's theorem gives

$$(4.10) \quad \Omega_0 = \int_{K_0} (1 + \|z\|^2)^{-\frac{n}{2}} dz \leq \int_{P_BK_0} \int_{U_B} (1 + \|x\|^2 + \|y\|^2)^{-\frac{n}{2}} dy dx.$$

Recall the following Laplace transform identity:

$$(4.11) \quad s^{-\frac{n}{2}} = \Gamma\left(\frac{n}{2}\right)^{-1} \int_0^\infty \tau^{\frac{n}{2}-1} e^{-s\tau} d\tau.$$

Consider any $x \in P_B K_0$. We apply (4.11) and find

$$\begin{aligned} \int_{U_B} (1 + \|x\|^2 + \|y\|^2)^{-\frac{n}{2}} dy &= \Gamma\left(\frac{n}{2}\right)^{-1} \int_0^\infty \int_{\mathbb{R}^{\ell_B}} \tau^{\frac{n}{2}-1} e^{-(1+\|x\|^2+\|y\|^2)\tau} dy d\tau \\ &= \Gamma\left(\frac{n}{2}\right)^{-1} \int_0^\infty \tau^{\frac{n}{2}-1} e^{-(1+\|x\|^2)\tau} \int_{\mathbb{R}^{\ell_B}} e^{-\tau\|y\|^2} dy d\tau \\ &= \Gamma\left(\frac{n}{2}\right)^{-1} \int_0^\infty \tau^{\frac{n-\ell_B}{2}-1} e^{-(1+\|x\|^2)\tau} d\tau \int_{\mathbb{R}^{\ell_B}} e^{-\|y\|^2} dy. \end{aligned}$$

Using the Laplace transform identity (4.11) once again and integrating the Gaussian yields

$$\int_{\mathbb{R}^{\ell_B}} (1 + \|x\|^2 + \|y\|^2)^{-\frac{n}{2}} dy = \pi^{\frac{\ell_B}{2}} \Gamma\left(\frac{n}{2}\right)^{-1} \Gamma\left(\frac{n-\ell_B}{2}\right) (1 + \|x\|^2)^{-\frac{n-\ell_B}{2}}.$$

With $n - \ell_B = \ell_A + 1$, this results in

$$(4.12) \quad \Omega_0 \leq \pi^{\frac{\ell_B}{2}} \Gamma\left(\frac{n}{2}\right)^{-1} \Gamma\left(\frac{\ell_A+1}{2}\right) \mathcal{H}^{\ell_A}(P_B K_0).$$

For the other solid angle Ω_n , a fully analogous integration argument with the two index sets switched yields

$$(4.13) \quad \Omega_n \leq \pi^{\frac{\ell_A}{2}} \Gamma\left(\frac{n}{2}\right)^{-1} \Gamma\left(\frac{\ell_B+1}{2}\right) \mathcal{H}^{\ell_B}(P_A K_n).$$

We combine (4.6), (4.7), (4.8), (4.9), (4.12), and (4.13) and recall $\ell_A + \ell_B = n - 1$. Now,

$$(4.14) \quad \Omega_S^2 \leq \Omega_0 \Omega_n \leq \frac{\pi^{\frac{n-1}{2}} \Gamma\left(\frac{\ell_A+1}{2}\right) \Gamma\left(\frac{\ell_B+1}{2}\right) \Gamma\left(\frac{n}{2}\right)^{-2}}{\prod_{i \in I_A} \theta_i \prod_{i \in I_B} (1 - \theta_i)} \left(\frac{(n-1)!}{\ell_A! \ell_B!} \right)^2 \frac{V_F^2}{V_A V_B}, \quad \frac{V_F^2}{V_A V_B} \leq \frac{\ell_A! \ell_B!}{(n-1)!} n V_S.$$

An improvement involves Theorem 3.1 and concerns the extremal cases of the index partitions. If $\ell_A = 0$, then $P_A K_n = K_n$ and

$$\Omega_S^2 \leq \Omega_S \Omega_n \leq \frac{|\hat{S}^{n-1}|}{2n+2} \mathcal{H}^{n-1}(K_n) = \frac{|\hat{S}^{n-1}|}{2n+2} \prod_{1 \leq i \leq n-1} (1 - \theta_i)^{-1} V_F = \frac{\pi^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)^{-1} (n+1)^{-1} n}{\prod_{1 \leq i \leq n-1} (1 - \theta_i)} V_S.$$

If $\ell_A = n - 1$, then $P_B K_0 = K_0$ and

$$\Omega_S^2 \leq \frac{\pi^{\frac{n}{2}} \Gamma\left(\frac{n}{2}\right)^{-1} (n+1)^{-1} n}{\prod_{1 \leq i \leq n-1} \theta_i} V_S.$$

We sum over all the possible partitions (4.5). Since the coefficients only depend on the cardinalities of the index sets, we arrive at the upper bound

$$\Omega_S^2 \leq C_n V_S, \quad C_n := n \pi^{\frac{n-1}{2}} \sum_{\ell=1}^{n-2} \binom{n-1}{\ell}^2 \frac{\Gamma\left(\frac{\ell+1}{2}\right) \Gamma\left(\frac{n-\ell}{2}\right)}{\Gamma\left(\frac{n}{2}\right)^2} + \frac{2n \pi^{\frac{n}{2}}}{(n+1) \Gamma\left(\frac{n}{2}\right)}.$$

Since $\delta_S = 1$, we have $V_S = \mu_S$ and thus (4.2) follows. For general simplices with arbitrary diameters, the desired result follows by scaling invariance of μ_S and of the solid angles. $\square$

Remark 4.5. A simplified, albeit rougher, estimate is of interest. With the notation as in the proof of Theorem 4.4, we consider the particular partition

$$I_A := \{1 \leq i \leq n-1 \mid \theta_i > \frac{1}{2}\}, \quad I_B := \{1 \leq i \leq n-1 \mid \theta_i \leq \frac{1}{2}\}.$$

Using this partition in (4.14), this delivers the upper bound

$$(4.15) \quad \Omega_S^2 \leq \Omega_0 \Omega_n \leq 2^{n-1} \pi^{\frac{n-1}{2}} \Gamma\left(\frac{\ell_A+1}{2}\right) \Gamma\left(\frac{\ell_B+1}{2}\right) \Gamma\left(\frac{n}{2}\right)^{-2} \frac{n!}{\ell_A! \ell_B!} V_S.$$

By taking the maximum over $0 \leq \ell_A \leq n-1$, the resulting constant only depends on n . The preceding proof can be interpreted as a refinement of this initial rough estimate.

Example 4.6. The following example illustrates the optimality of the quadratic exponent in Theorem 4.4 for simplices of any dimension n . Let $\epsilon > 0$. The vertices

$$v_0 = 0, \quad v_1 = \epsilon e_1, \quad \dots \quad v_{n-2} = \epsilon e_{n-2}, \quad v_{n-1} = e_{n-1} - \epsilon^{n-2} e_n, \quad v_n = e_{n-1} + \epsilon^{n-2} e_n$$

define an n -dimensional simplex. Simple direct computations show that

$$\delta_S = \sqrt{1 + \epsilon^2 + \epsilon^{2(n-2)}}, \quad V_S = \frac{2}{n!} \epsilon^{2(n-2)}, \quad \mu_S = \frac{n!(1 + \epsilon^2 + \epsilon^{2(n-2)})^{\frac{n}{2}}}{2\epsilon^{2(n-2)}}$$

By symmetry, the solid angle $\Omega_0(S)$ equals a fraction of the semisphere area, $\zeta|\hat{S}^{n-1}|/4$, where $\zeta = 2 \arctan(\epsilon^{n-2})/\pi$. For $\epsilon > 0$ small enough,

$$\frac{|\hat{S}^{n-1}|}{8} \epsilon^{n-2} \leq \Omega_0 \leq \frac{|\hat{S}^{n-1}|}{2\pi} \epsilon^{n-2}.$$

By symmetry, too, $\Omega_1(S) = \dots = \Omega_{n-2}(S)$ and $\Omega_{n-1}(S) = \Omega_n(S)$. Let

$$M_1 := \begin{pmatrix} -\epsilon & -\epsilon & \dots & -\epsilon & -\epsilon & -\epsilon \\ & \epsilon & & & & \\ & & \ddots & & & \\ & & & \epsilon & & \\ & & & & 1 & 1 \\ & & & & -\epsilon^{n-2} & \epsilon^{n-2} \end{pmatrix}, \quad M_n := \begin{pmatrix} & & & \epsilon & & \\ & & & & \ddots & \\ & & & & & \epsilon \\ -1 & -1 & \dots & -1 & & \\ -\epsilon^{n-2} & -\epsilon^{n-2} & \dots & -\epsilon^{n-2} & -2\epsilon^{n-2} & \end{pmatrix}$$

be the matrices whose columns are the edges of the simplex S that emanate from the vertices v_0 and v_n . For any $x \in \mathbb{R}^n$ with positive entries, expanding the norm of $M_1 x$ gives

$$\|M_1 x\|^2 = \epsilon^2 (\sum_{i=1}^n x_i)^2 + \epsilon^2 \sum_{i=2}^{n-2} x_i^2 + (x_{n-1} + x_n)^2 + \epsilon^{2(n-2)} (x_{n-1} - x_n)^2.$$

Assuming that $\epsilon > 0$ is small, the following simple estimates hold:

$$\begin{aligned} \|M_1 x\|^2 &\leq 2\epsilon^2 (\sum_{i=1}^{n-2} x_i)^2 + \epsilon^2 \sum_{i=2}^{n-2} x_i^2 + (1 + 2\epsilon^2 + \epsilon^{2(n-2)}) (x_{n-1} + x_n)^2 \\ &\leq \epsilon^2 (2n - 4 + \epsilon^2) (\sum_{i=1}^{n-2} x_i^2) + 2(1 + \epsilon^2 + \epsilon^{2(n-2)}) (x_{n-1}^2 + x_n^2) \\ \|M_1 x\|^2 &\geq \epsilon^2 (\sum_{i=1}^{n-2} x_i)^2 + (1 + \epsilon^{2(n-2)}) (x_{n-1}^2 + x_n^2) + 2(1 - \epsilon^{2(n-2)}) x_{n-1} x_n \\ &\geq \epsilon^2 (\sum_{i=1}^{n-2} x_i^2) + (1 + \epsilon^{2(n-2)}) (x_{n-1}^2 + x_n^2) \end{aligned}$$

Applying the lower and upper bound in the representation (3.3) of the solid angles as orthant integral and performing a substitution, together with $n!V_S = 2\epsilon^{n-2}$, yields

$$(1 - \epsilon) \frac{\epsilon^{n-2}}{2^n (2n - 4)^{\frac{n-2}{2}}} \leq \Omega_1(S) \leq \frac{\epsilon^{n-2}}{2^{n-1}}$$

Similarly, the norm of $M_n x$ expands like

$$\|M_n x\|^2 = \epsilon^2 \sum_{i=2}^{n-1} x_i^2 + (\sum_{i=1}^{n-1} x_i)^2 + \epsilon^{2(n-2)} (\sum_{i=1}^{n-1} x_i + 2x_n)^2,$$

and we derive the estimates

$$\begin{aligned} \|M_n x\|^2 &\leq \epsilon^2 \sum_{i=2}^{n-1} x_i^2 + (1 + 2\epsilon^{2(n-2)}) (\sum_{i=1}^{n-1} x_i)^2 + 8\epsilon^{2(n-2)} x_n^2 \\ &\leq (2 + \epsilon^2 + 4\epsilon^{2(n-2)}) (\sum_{i=1}^{n-1} x_i^2) + 8\epsilon^{2(n-2)} x_n^2 \\ \|M_n x\|^2 &\geq \sum_{i=1}^{n-1} x_i^2 + \epsilon^{2(n-2)} 4x_n^2. \end{aligned}$$

Again applying the lower and upper bound in the representation (3.3) results in the following lower and upper bound for the solid angle:

$$(1 - \epsilon) 2^{-\frac{3n}{2}} \epsilon^{n-2} \leq \Omega_n(S) \leq 2^{-n} \epsilon^{n-2}.$$

We conclude that there exist constants $c_n, C_n > 0$, depending only on n , such that

$$c_n \Omega_S^2 \leq V_S \leq C_n \Omega_S^2.$$

This illustrates that the quadratic exponent 2 is optimal.

We finish this discussion with low-dimensional examples. Triangles admit an improved bound where the fullness is bounded not only from above but also from below by multiples of the minimum (solid) angle. Tetrahedral minimum angle conditions were studied by Liu and Joe [19], and both Theorem 4.1 and Theorem 4.4 for $n = 3$ yield improved bounds. Lastly, the minimum solid angle condition for pentatopes is illustrated.

Example 4.7. Let S be a triangle with angles $\alpha, \beta, \gamma \in (0, \pi)$ and opposite side lengths $a, b, c > 0$, respectively. Assume that α equals the minimal angle and that c equals the diameter. Using the law of sines,

$$2V_S = ab \sin(\gamma) = c^2 \frac{\sin(\alpha) \sin(\beta)}{\sin(\gamma)} = c^2 \sin(\alpha) \frac{\sin(\beta)}{\sin(\alpha + \beta)}.$$

Since $\alpha \leq \beta \leq \pi - \alpha$ and $3\alpha \leq \pi$, we have $\sin(\alpha) \leq \sin(\beta)$ and $\cos(\beta) \leq \cos(\alpha)$. From $b \leq c$ we get $\sin(\beta) \leq \sin(\gamma)$, and since $\beta + \gamma < \pi$ we get $\beta \leq \gamma$, hence $\cos(\beta) \geq 0$. By the angle sum identity,

$$\frac{\sin(\alpha + \beta)}{\sin(\beta)} = \frac{\sin(\alpha) \cos(\beta) + \cos(\alpha) \sin(\beta)}{\sin(\beta)} \leq \cos(\beta) + \cos(\alpha) \leq 2 \cos(\alpha).$$

With $\alpha \leq \tan(\alpha)$ and $\sin(\alpha) \leq \alpha$, we find

$$\frac{1}{4}\alpha \leq \frac{1}{4}\tan(\alpha) \leq \frac{V_S}{\delta_S^2} \leq \frac{1}{2}\sin(\alpha) \leq \frac{1}{2}\alpha.$$

This proves the minimum angle of a triangle to be a shape measure comparable to the fullness. Using Theorem 4.4 and $\alpha \leq \pi/3$, we get the weaker estimates

$$(4.16) \quad \frac{3}{4\pi}\alpha^2 \leq \mu_S \leq \frac{1}{2}\alpha, \quad \frac{3}{4\pi}\alpha^2 \leq \frac{1}{4}\alpha.$$

Thus, Theorem 4.4 recovers natural inequalities relating the triangle fullness and minimum angle.

Example 4.8. We consider a tetrahedron S with vertices v_0, v_1, v_2, v_3 . Liu and Joe [19] compare shape measures of tetrahedra, including the minimum solid angle Ω_S and the *mean ratio* η_S , where [19, (3.4)]

$$\eta_S := 12(3V_S)^{\frac{2}{3}} \left(\sum_{0 \leq i < j \leq 3} \|v_i - v_j\|^2 \right)^{-1}.$$

This implies a comparison between the fullness and the minimum solid angle of tetrahedra, as we now show. Without loss of generality, $\delta_S = \|v_0 - v_1\|$. Whereas $\|v_2 - v_3\|$ can be arbitrarily small,

$$\|v_0 - v_2\|^2 + \|v_2 - v_1\|^2 + \|v_0 - v_3\|^2 + \|v_3 - v_1\|^2 \geq \delta_S^2.$$

Therefore, the fullness and the mean ratio satisfy the comparison

$$2^{\frac{3}{2}} 3\mu_S \leq \eta_S^{\frac{3}{2}} \leq 6^{\frac{3}{2}} 3\mu_S.$$

Define $\sigma_S := \sin(\Omega_S/2)$. By Liu and Joe's comparison result [19, Theorem 5.2],

$$\frac{1}{16}\eta_S^{\frac{3}{2}} \leq \sigma_S \leq 8^{\frac{1}{4}}\eta_S^{\frac{3}{4}}.$$

The sum of the four solid angles in S is at most 2π , thus $\Omega_S \leq \pi/2$. The inequalities $\frac{\sqrt{2}}{\pi}\Omega_S \leq \sigma_S \leq \frac{1}{2}\Omega_S$ follow. The combination of these findings gives

$$(4.17) \quad 3\frac{\sqrt{2}}{4}\mu_S \leq \Omega_S, \quad \Omega_S^2 \leq 36\sqrt{3}\pi^2\mu_S.$$

We use Theorem 4.4 with $n = 3$ and the fact $\Gamma(\frac{3}{2}) = \sqrt{\pi}/2$ to find the stronger estimates

$$(4.18) \quad 3\mu_S \leq \Omega_S, \quad \Omega_S^2 \leq (48 + 3\pi)\mu_S.$$

In addition, Theorem 4.1 further improves the constant and yields

$$(4.19) \quad 3\mu_S \leq \Omega_S, \quad \Omega_S^2 \leq (12 + 3\pi)\mu_S.$$

In particular, while a comparison between fullness and minimum solid angle of a tetrahedron already follows from known results, this manuscript yields tighter constants for this comparison of shape measures.

Example 4.9. If $S \subset \mathbb{R}^4$ is a 4-simplex (pentatope), then Theorem 4.4 yields

$$(4.20) \quad 4\mu_S \leq \Omega_S, \quad \Omega_S^2 \leq \frac{188}{5}\pi^2\mu_S.$$

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